

EviSeq: Source-Grounded Summarization with Composed Pretrained Encoders and Causal Decoders

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Abstract

Long-document summarization asks a model to shorten a source without losing the evidence that makes it useful. This becomes difficult when the source encoder and causal decoder come from separate pretrained checkpoints: matching their widths makes the tensors compatible but does not determine how source information should reach the generator. We present EviSeq, a source-grounded summarization system built around an Adaptive Full-Memory Residual interface. AFMR retains every visible source position, separates adaptable retrieval keys from aligned final-state values, and learns bounded source-priority adjustments. Copied cross-attention lets each decoder layer read this memory, while an offset-based copy route provides access to source wording and numbers. The model is trained with a single gold-token likelihood objective and evaluated with deterministic decoding. We compare it with fine-tuned decoder-only models and an encoder-decoder reference across long-document domains, examining source consistency, summary quality, component contributions, and encoder effects. Implementation tests support the executable design; held-out predictions and empirical comparisons remain pending.

1 Introduction

Long-document abstractive summarization turns a lengthy source into a shorter text. For scientific, biomedical, book, and government documents, a useful summary must preserve entities, numbers, and relations that the source supports. Factual support is therefore different from lexical overlap: a fluent summary can resemble its reference while changing a quantity or relation. Prior work has documented this gap in abstractive summarization (Maynez et al., 2020).

Most sequence-to-sequence models learn source encoding and target generation through one interface, with the decoder attending to encoder states

(Sutskever et al., 2014; Bahdanau et al., 2015; Vaswani et al., 2017). T5, BART, and mT5 pre-train this interface jointly (Raffel et al., 2020; Lewis et al., 2020; Xue et al., 2021). A different problem appears when an embedding-oriented encoder and a causal decoder are reused from separate checkpoints. Their hidden widths, representation coordinates, tokenizers, and pretraining objectives may all differ. A projection can make the widths compatible, but it does not determine which source representations, positions, or lexical forms should guide each generated token.

This composition creates two related problems. The interface must keep every visible source position available while assigning useful priority across encoder depth, feature channels, and source spans. It must also expose source words and other lexical forms without allowing a separate retrieval objective to dictate generation. Long-document attention and copying address parts of these problems, but they do not specify how to allocate information between independently pretrained representation spaces (Cohan et al., 2018; See et al., 2017). We frame that allocation as an interface problem and test it with the same generation graph and matched comparisons.

We study this problem with EviSeq and its proposed Adaptive Full-Memory Residual (AFMR) interface. AFMR keeps a full retrieval tensor for every visible source position alongside an aligned final-state value tensor, then learns small adjustments conditioned on the source and decoder budgets. These adjustments act along three axes. A depth route combines the final encoder state with the last four hidden-state taps around a final-state anchor. A cross-space feature route adapts the encoder representation to the decoder width while leaving the main projection path available. A multi-scale route builds a shared additive prior from overlapping source-token windows of lengths 32, 128, and 512 in the base profile. Every

decoder layer reads these tensors through a copied cross-attention block: refined representations form the keys, and final-state representations supply the values. A grounded-copy route aligns source and decoder-token offsets, combines lexical and contextual keys, and marginalizes repeated source token IDs. The model keeps all visible source positions and needs no evidence labels at inference.

The graph uses the gold-token mixture cross-entropy that defines generation. Training begins with one interface warm-up epoch and then moves to full fine-tuning; official generation uses one beam with sampling disabled. We apply the same preprocessing, splits, source and target budgets, and training budgets to PubMed, arXiv, BookSum, and GovReport. The comparisons include fine-tuned decoder-only controls and fine-tuned T5Gemma2. Implementation and unit tests can verify tensor, masking, gradient, checkpoint, and inference invariants, but the four-dataset predictions and quality measurements are not available at this stage. AFMR is consequently a testable hypothesis, and we make no claim of improved ROUGE or factuality before the registered runs.

Across PubMed, arXiv, BookSum, and GovReport, we ask four linked questions:

- **RQ1 (factual-support signal):** Does EviSeq reduce a source-unsupported-content signal relative to fine-tuned decoder-only LLMs on all four datasets? We use native AlignScore consistency and its direction-transformed $H = 1 - C$ value for the automatic comparison; a claim about typed entity, number, or relation errors requires a separate adjudicated audit.
- **RQ2 (quality):** How does EviSeq change ROUGE-1/2/L and BERTScore relative to the same decoder-only controls and fine-tuned T5Gemma2 under matched records, budgets, preprocessing, and decoding? Each model’s native input serialization is documented for all four datasets.
- **RQ3 (mechanism):** Which of the encoder, AFMR bridge, and grounded-copy route contributes to the observed behavior? We run no-encoder, no-bridge, and no-grounded-copy controls on all four datasets with a fixed decoder and matched training budget.
- **RQ4 (portability):** Does the same interface

transfer across a bidirectional embedding encoder, a causal encoder, and a larger embedding encoder, and how do directionality and capacity affect quality and factuality? We use PPLX-Embed v1 0.6B as the anchor and test Qwen3 causal encoder, Qwen3-Embedding-0.6B, and Nemotron-3-Embed-1B variants on all four datasets.

Together, the four questions separate source support, summary quality, component contribution, and encoder portability. RQ1 tests whether the interface changes an automatic source-support signal. RQ2 checks whether that change comes with different reference overlap or semantic similarity. RQ3 identifies the routes that contribute to the observed benchmark behavior, while RQ4 asks whether the behavior persists across encoders with different directionality and capacity. For RQ1, higher native AlignScore consistency is better. We report $H = 1 - C$, where $C = \text{AlignScore}_{\text{nli_sp}}$, only as a direction-transformed hallucination-oriented score; lower is better, and H is not a calibrated probability. RQ2 uses ROUGE-1/2/L F1 and BERTScore-F1 as complementary measures of overlap and semantic similarity. All interpretations are conditional on the held-out predictions and paired uncertainty intervals.

This paper contributes three things. First, it formulates AFMR as a constrained full-memory interface for composing a pretrained encoder with a causal decoder, with key-side adaptation separated from final-state values. Second, it combines controller-conditioned depth, cross-space feature, and multi-scale span adjustments with a grounded-copy route while keeping one gold-token generation objective. Third, it defines a reproducible four-dataset evaluation contract that separates source-support signal, overlap quality, component contribution, and encoder portability, and records the diagnostics and uncertainty estimates needed to interpret them. The interface is experimentally falsifiable; these contributions do not imply that any component is necessary or that the complete system is superior before the planned comparisons.

2 Related Work

Encoder–decoder interfaces and checkpoint composition. Sequence-to-sequence learning and attention established the division between

source encoding and autoregressive generation (Sutskever et al., 2014; Bahdanau et al., 2015; Vaswani et al., 2017). T5, BART, and mT5 learn this interface during joint pretraining, so their source and target representations are optimized together (Raffel et al., 2020; Lewis et al., 2020; Xue et al., 2021). Other work reuses decoder-only checkpoints for representation learning, including LLM2Vec and Generative Representational Instruction Tuning (BehnamGhader et al., 2024; Muennighoff et al., 2024). LaMaTE uses a language-model encoder in a translation system, while Qwen3 Embedding provides an embedding-oriented model family (Luo et al., 2025; Zhang et al., 2025b). T5Gemma2 is a close reference for arranging modern language-model components as an encoder–decoder system (Zhang et al., 2025a). AFMR shares the goal of reusing pretrained components and the decoder’s cross-attention interface, but addresses the allocation problem that remains after composing checkpoints that were not released as a matched encoder–decoder model. We claim no novelty for generic checkpoint reuse or cross-attention itself.

Long-document source access. Long-document summarization has used hierarchical and discourse-aware attention to represent documents at several granularities (Cohan et al., 2018). These methods motivate looking at both local and broader source regions, but they do not decide how a composed model should turn regional preference into decoder retrieval. AFMR uses overlapping token windows at several scales to build a soft source prior. The decoder still receives retrieval and aligned value tensors for every visible source token. This leaves hard selection, sentence slots, and source compression outside the proposed interface and allows direct comparison with no-bridge and matched-encoder controls.

Copying and source coverage. Pointer-generator networks mix vocabulary generation with an attention-based copy distribution, while coverage mechanisms can discourage repeated attention to the same source content (See et al., 2017). AFMR uses this generation–copy decomposition for independently tokenized components. Character-offset overlap supplies a sparse source-to-decoder alignment; contextual and lexical features form the copy keys; and repeated source occurrences are summed by decoder token ID. The

copy route is trained with the same token-level mixture likelihood as the language-model path. AFMR has no separate coverage loss and does not treat copying as factual verification: a copied source token can still appear in the wrong relation or role.

Factuality and summary evaluation. ROUGE measures overlap with a reference (Lin, 2004), whereas BERTScore measures contextual semantic similarity between generated and reference texts (Zhang et al., 2020). Neither metric directly guarantees factuality, as prior work on faithfulness and evaluation has shown (Maynez et al., 2020; Fabbri et al., 2021). We therefore use AlignScore as the primary automatic source-consistency signal, report its native direction, and show $H = 1 - C$ for $C = \text{AlignScore}_{\text{nli_sp}}$ only as a lower-is-better diagnostic (Zha et al., 2023). AlignScore is still a learned proxy and may miss cross-sentence or domain-specific errors; BERTScore is not a hallucination detector. Our evaluation pairs these metrics with prediction diagnostics and, where available, a manual factual-error audit.

These lines of work provide the pieces used by AFMR: source–target attention, multi-scale source reasoning, and vocabulary–copy mixtures. AFMR brings them together at a full-memory boundary, adds bounded controller-conditioned residuals for encoder depth, cross-space features, and source-span priority, and trains the graph with gold-token cross-entropy. RQ3 tests whether these additions matter with a fixed decoder and matched budget; RQ4 tests transfer across the specified encoder families. The literature motivates the design, but only the locked four-dataset experiments can support claims about quality, factual support, component contribution, or portability.

3 Method

3.1 Problem setting and design goal

Let $x = (x_1, \dots, x_S)$ denote a source document and $y = (y_1, \dots, y_T)$ its reference summary. The system pairs a pretrained source encoder with a pretrained causal decoder. Their tokenizers, hidden widths, and pretraining objectives may differ. The encoder returns a final hidden sequence $H^{(L)} \in \mathbb{R}^{S \times d_e}$ together with a small set of intermediate sequences. At each step, the decoder predicts y_t from the causal prefix $y_{<t}$ and source memory represented in its width d_d .

Two constraints shape the interface. First, every source position that survives tokenization and the length limit must remain available to cross-attention; hard selection could discard an entity or number that the summary needs later. Second, the interface must provide a controllable retrieval signal without changing the content representation so much that the pretrained vocabulary head or copy route loses its lexical anchor. EviSeq meets both constraints with the AFMR bridge. The bridge returns a full source tensor for retrieval, an aligned final-state tensor for values, and an additive source prior. The two tensors use the same source positions. The design therefore has no pruned or separately selected source bank and no target-derived evidence representation.

3.2 AFMR interface

Controller. The bridge summarizes the visible source and the fixed decoder instruction separately. Let $C \in \{0, 1\}^S$ mark source content and let $M_p \in \{0, 1\}^{S_p}$ mark prompt tokens. We compute

$$\begin{aligned}\bar{h} &= \text{MaskedMean}(H^{(L)}, C), \\ \bar{p} &= \text{MaskedMean}(P, M_p).\end{aligned}\quad (1)$$

Here, P comes from the decoder prompt-token embeddings. We RMS-normalize the two means separately, project them into a controller vector $c \in \mathbb{R}^{d_c}$, and add three quantities available at inference: the number $N = \sum_i C_i$ of visible content tokens, the requested output budget K , and their ratio. Let $K_+ = \max(K, 0)$. The base profile uses $d_c = 256$. Its inference-only budget vector is

$$\phi(N, K) = \begin{bmatrix} \log(1 + N) \\ \log(1 + K_+) \\ K_+ / \max(N, 1) \end{bmatrix}.$$

The controller vector is then

$$\begin{aligned}c &= \text{RMSNorm}(W_h \text{RMSNorm}(\bar{h}) \\ &\quad + W_p \text{RMSNorm}(\bar{p}) \\ &\quad + \text{MLP}(\phi(N, K))).\end{aligned}\quad (2)$$

The reference summary and its length never enter the construction of c .

Token-wise depth adaptation. The default bridge exposes the final four encoder states. For source position i and tap j , let $\tilde{H}_{i,j}$ denote the independently RMS-normalized state. A shared token scorer combines each tap with a controller

preference:

$$\begin{aligned}\pi_{i,j} &= \text{softmax}_j \left(u_d^\top \tilde{H}_{i,j} + (W_d c)_j \right), \\ D_i &= \sum_{j=1}^4 \pi_{i,j} \tilde{H}_{i,j}.\end{aligned}\quad (3)$$

The softmax acts over depth for each source token, not over source positions. We add a low-rank correction around the final-state anchor:

$$H_i^* = H_i^{(L)} + g_d(c) B_d \text{SiLU} \left(A_d [D_i - \tilde{H}_{i,4}] \right), \quad (4)$$

Here A_d has the base-profile rank 128, B_d starts at zero, and $g_d(c) = 0.15 \sigma(\cdot)$ is initialized at 0.02. Because the output factor starts at zero, the initial memory follows the final-state path. Gradients can still reach the residual and, after an update, the depth scorer and router.

Cross-space feature adaptation and value anchor. The base map Π is the identity when $d_e = d_d$; otherwise it is a trainable orthogonally initialized projection. The retrieval memory is

$$\begin{aligned}Z_i &= \Pi(H_i^*) + G_f(c) \odot B_f \text{SiLU} \\ &\quad (A_f \text{RMSNorm}(H_i^*)),\end{aligned}\quad (5)$$

Here A_f and B_f form the base-profile rank-256 correction. The feature-wise gate $G_f(c) = 0.20 \sigma(\cdot)$ starts at 0.02, and its output factor also starts at zero. The separate value anchor is

$$V_i^0 = \Pi(H_i^{(L)}). \quad (6)$$

This lets Z learn a retrieval-friendly representation while V^0 keeps the final encoder content as its anchor. The separation is a graph constraint, not an assertion that the values stay fixed during full fine-tuning. Shared encoder and projection parameters still receive gradients through both paths.

Soft multi-scale source prior. AFMR keeps every valid position and builds a relative prior from overlapping content-token windows. The base profile uses widths $\mathcal{W} = \{32, 128, 512\}$ with 50% overlap. A task recipe may register a different capacity profile, but the run manifest records the resolved architecture and dimensions. For each window, the controller and a scale embedding score a masked mean of Z . We center these scores within each document and scale, lift them to tokens with

overlap-add, and mix the available scales. The focus strength and temperature are bounded by

$$\begin{aligned}\lambda(c) &= 1.0 \sigma(w_\lambda^\top c + a_\lambda), \\ \tau(c) &= 0.5 + 1.5 \sigma(w_\tau^\top c + a_\tau).\end{aligned}\quad (7)$$

The resulting score is $r_i = \lambda(c) \tanh(s_i/\tau(c))$. For content positions $\mathcal{C}$, we center the source bias with its log-mean-exp:

$$\begin{aligned}b_i &= r_i - \log\left(\frac{1}{|\mathcal{C}|} \sum_{k \in \mathcal{C}} \exp r_k\right), \quad i \in \mathcal{C}, \\ b_i &= 0 \quad i \notin \mathcal{C}.\end{aligned}\quad (8)$$

Cross-attention masks padding separately. The prior changes relative source preference, but it neither prunes the memory nor fixes the final attention mass. All reductions in the focus path use FP32 before the bias is cast to the attention dtype.

3.3 Decoder source access and grounded copying

Copied cross-attention. Each causal decoder layer keeps its pretrained self-attention and applies a copied cross-attention block afterward. We initialize the query, key, value, output, and normalization projections from the corresponding self-attention modules. For layer ℓ and target position t ,

$$\begin{aligned}h'_{\ell,t} &= h_{\ell,t} + g_\ell \mathcal{A}_{\ell,t}, \\ g_\ell &= \sigma(\theta_\ell),\end{aligned}\quad (9)$$

Here $\mathcal{A}_{\ell,t}$ is multi-head attention with query $\text{RMSNorm}(h_{\ell,t})$, keys from Z , values from V^0 , additive bias b , and mask M . The gate g_ℓ starts at 0.10 and is bounded by 1.0. The attention logits therefore contain $QK(Z)^\top/\sqrt{d_h} + b$ together with the padding mask. All layers receive the same source prior, while each layer keeps its own query and output. During greedy generation, the implementation prepares the source keys and values once per layer and reuses them through the cache.

Grounded copy and mixture likelihood. The copy head uses fast-tokenizer character offsets to align visible source spans with decoder vocabulary tokens. It pools contextual source keys, combines them with lexical decoder embeddings, and computes decoder-query attention over eligible aligned source-token rows $j \in \mathcal{J}$. The same offset weights pool the source prior into $\tilde{b}_j$, giving

$$A_{t,j}^{\text{copy}} = \text{softmax}_{j \in \mathcal{J}} \left(\frac{q_t^\top k_j}{\sqrt{d_k}} + \tilde{b}_j \right). \quad (10)$$

Repeated source occurrences are then marginalized:

$$P_{\text{copy}}(v | t) = \sum_{j \in \mathcal{J}: \text{tok}_j=v} A_{t,j}^{\text{copy}}. \quad (11)$$

The learned gate ρ_t mixes this distribution with the vocabulary head applied to the decoder hidden state:

$$\begin{aligned}P(y_t = v | y_{<t}, x) &= \rho_t P_{\text{copy}}(v | t) \\ &\quad + (1 - \rho_t) P_{\text{LM}}(v | h_t).\end{aligned}\quad (12)$$

The copy gate starts at 0.05. When no eligible source token is aligned, the model uses the LM logits exactly. Alignment reads only the source and tokenizers; it never reads the reference summary or evidence labels.

3.4 Training, gradient flow, and inference parity

Training minimizes the token-level negative log likelihood of the mixture:

$$\mathcal{L}_{\text{CE}} = - \sum_{t=1}^T \log P(y_t | y_{<t}, x). \quad (13)$$

The graph uses no contrastive, ranking, distillation, pseudo-reference, or self-improvement objective. During interface warm-up, the pretrained backbones stay frozen while the bridge, copied cross-attention, and grounded copy parameters learn from this loss. Full fine-tuning then updates the encoder and decoder jointly. Parameters, gradients, and AdamW states remain in FP32, while CUDA BF16 autocast handles the heavy operations. The implementation clips the global norm at the configured training threshold and computes cross-entropy in chunks, so it need not retain a full vocabulary-logit tensor.

The live gradient route is

$$\begin{aligned}\mathcal{L}_{\text{CE}} &\rightarrow \{P_{\text{LM}}, P_{\text{copy}}\} \rightarrow \{h_t, Z, V^0, b\} \\ &\rightarrow \{\text{decoder, AFMR, encoder}\}.\end{aligned}$$

No detach is placed on Z, V^0, b , the depth mixture, or the controller. The value-anchor separation constrains the direct route from the retrieval residual to values, while encoder gradients still pass through both paths.

Teacher forcing and inference use the same encoder, prompt embeddings, requested budget, AFMR bridge, source memory, source prior,

and copy alignment. Official evaluation uses one beam, no sampling, and autoregressive argmax with `temperature=0`, `top_k=0`, and `top_p=1`. These fields remain available for a separate candidate-generation path but do not affect training. The encoder and bridge run once per example; cached cross-attention states then advance with the generated prefix.

4 Experiments

4.1 Research questions and comparisons

We evaluate the same questions on PubMed, arXiv, BookSum, and GovReport. RQ1 compares EviSeq with fine-tuned decoder-only controls on source support. RQ2 compares those systems and T5Gemma2 on overlap and semantic quality. For RQ3, we train three ablations from the pre-trained initialization with the decoder held fixed: (i) no encoder, implemented as a decoder-only source-prefix control; (ii) no AFMR bridge, using the direct final-state projection with zero source prior; and (iii) no grounded copy, using the complete AFMR and decoder path with only the vocabulary head. RQ4 keeps the decoder and interface fixed while replacing the anchor encoder PPLX-Embed v1 0.6B with Qwen3 causal encoder, Qwen3-Embedding-0.6B, and Nemotron-3-Embed-1B. Each RQ covers all four datasets. A missing manifest or failed run stays missing rather than being replaced with a score from another split.

The decoder-only suite contains Qwen3-0.6B, Qwen3-4B, Qwen3-8B, Llama 3.2 3B Instruct, Llama 3 8B Instruct, Nemotron 3B, and Nemotron 8B. We fine-tune each model locally on the same source–target records. We also fine-tune T5Gemma2 on every dataset and use it as the encoder–decoder reference. For each ablation and encoder-transfer row, all non-target components and optimization budgets remain fixed; the full baseline suite is used in the main RQ1/RQ2 tables.

4.2 Datasets and preprocessing

PubMed pairs biomedical articles with abstracts, and arXiv pairs scientific papers with abstracts. BookSum contains long narrative documents with human-written summaries at multiple granularities (Kryscinski et al., 2022), whereas GovReport contains long U.S. government reports with expert-written summaries (Huang et al., 2021). We will record each dataset’s official split and task variant in a manifest. The repository currently pro-

vides canonical JSONL preparation for PubMed and arXiv; the BookSum and GovReport manifests and field mappings are still pending, so this draft cannot report their scores.

We normalize each record to `id`, `text`, and `summary`. The encoder receives a fixed task prefix, the source, and a tokenizer-specific special sequence. The content mask removes the prefix, special tokens, and padding from the depth and focus calculations. Each task uses one fixed decoder instruction, tokenized once and excluded from the supervised labels. Every model in a dataset uses the same source and target length limits, and we report truncation counts. The model receives no reference text at inference.

4.3 Metrics and statistical analysis

We use ROUGE-1, ROUGE-2, and ROUGE-L F1 as the primary quality measures. One Perl ROUGE-1.5.5 wrapper and one preprocessing procedure serve every system (Lin, 2004). Python ROUGE-1.0.0 is kept for debugging only and is never mixed with the primary values. BERTScore-F1 is the secondary headline measure for semantic similarity; precision and recall are reported as diagnostics (Zhang et al., 2020). The evaluator loads its backbone from a local path and fails instead of downloading another model.

RQ1 uses the native consistency output of the local AlignScore `nli_sp` procedure. For example i , we pair source chunks with each generated claim sentence, keep the strongest support for each claim, and average the claim scores to obtain $C_i = \text{AlignScore}_{\text{nli_sp}}(x_i, y_i)$ (Zha et al., 2023). The dataset estimate is the mean of C_i , with higher native consistency preferred. For readability, we may also show $H_i = 1 - C_i$ and its dataset mean as a lower-is-better, direction-transformed hallucination-oriented value. This transform is neither a probability nor a calibrated percentage of hallucinated content. We report the model/checkpoint identifiers, maximum pair length, chunking, and sentence splitter for both AlignScore and BERTScore.

For every paired comparison, we will compute a 95% paired bootstrap interval over identical test IDs and report the number of examples. The dataset-level estimate is the mean over its examples. The secondary four-dataset macro estimate is the unweighted mean of the four dataset-level effects, with a dataset-aware interval. We report dataset rows first and leave the macro value

pending when any dataset is missing. Each research question has one pre-registered primary endpoint; other metrics and model contrasts are exploratory unless we specify a multiplicity adjustment. We will also record generated and reference lengths, length ratio, empty and truncated outputs, repeated-trigram rate, unique-output rate, source-copy rate, and, when available, a manually audited factual-error subset. These diagnostics show whether an apparent factuality gain could instead come from shorter or empty outputs.

4.4 Fairness, implementation, and reporting

Within a dataset, all main models use the same split IDs, source and target budgets, preprocessing, seed policy, optimizer family, number of effective training examples, and greedy decoding constraints. We record each model’s native source and decoder serialization explicitly; a shared task instruction does not make tokenization or chat-template behavior identical. The AFMR recipe uses interface warm-up followed by full fine-tuning. We match the number of optimizer updates to the corresponding baseline instead of inferring it from wall-clock time. The reported checkpoint is `last.pt`; test scores never select a best checkpoint. We record batch size and accumulation for each model and dataset so the effective batch is clear.

The experiment manifest will record local model paths or immutable revisions, tokenizer special IDs, prompts, source and target limits, seeds, total and trainable parameter counts, precision, software versions, GPU type, peak VRAM, training time, generation time, failed runs, and output-file checksums. A score cell is valid only when its prediction manifest and metric JSON agree on the dataset split, example IDs, checkpoint, tokenizer, scorer, and decoding settings. The four-dataset tables in Section 5 remain blank until these checks pass.

5 Results

Evidence status. At this stage, the evidence covers implementation behavior rather than generation quality. The local `src/eviseq_new` suite runs the value-anchored AFMR and grounded-copy graph on model-free fixtures. It checks source masking and alignment, cross-attention and cache behavior, cross-entropy gradient reachability, dense and chunked loss parity, check-

point round trips, warm-up and full-training paths, and greedy generation with prediction resumption. The source documentation also defines a smoke path that trains, saves, reloads, and evaluates the graph with tiny randomly initialized backbones. These checks show that the specified tensor contracts and runtime paths execute on controlled inputs. They do not yield a held-out comparison, a ROUGE-1/2/L value, a factuality estimate, a BERTScore value, or a hardware throughput benchmark.

The four-dataset fine-tuning runs and prediction files are not yet available. The fixed Perl ROUGE-1.5.5 scorer, AlignScore evaluator, and BERTScore evaluator therefore have no results to report. Dashes in Table 1 mean “not yet computed”; they do not mean zero, failure, or imputation. This draft makes no quality claim.

Evaluation contract. The primary comparison uses fine-tuned decoder-only controls: Qwen3-0.6B, Qwen3-4B, Qwen3-8B, Llama 3.2 3B Instruct, Llama 3 8B Instruct, Nemotron 3B, and Nemotron 8B. It also includes the fine-tuned encoder-decoder reference T5Gemma2. On each of PubMed, arXiv, BookSum, and GovReport, the runs share example IDs, fixed train/validation/test splits, source and target budgets, preprocessing, training-stage budgets, and the registered decoding procedure. We report native model serialization and prompt/chat-template behavior instead of assuming that they match. One fixed Perl ROUGE-1.5.5 wrapper supplies the primary ROUGE-1/2/L F1 values; Python ROUGE remains a diagnostic backend and is never combined with the paper scores. Each comparison will include a per-example paired bootstrap interval. A macro-average across the four datasets is a secondary summary. The prediction manifest will record output and reference lengths, truncation, empty outputs, repetition, source-copy rate, latency, peak memory, checkpoint identity, and failed or omitted runs.

RQ1: factual-support signal. RQ1 asks whether EviSeq changes an automatic source-support signal relative to the fine-tuned decoder-only controls on PubMed, arXiv, BookSum, and GovReport. Its primary quantity is native AlignScore consistency, where a higher value means greater source-summary consistency under the metric. We may also show $H = 1 - C$ for $C = \text{AlignScore}_{\text{nli_sp}}$ as a

Table 1: Pre-results reporting plan. Each outcome cell remains blank until the corresponding scorer-locked run and provenance check are complete.

Planned analysis	PubMed	arXiv	BookSum	GovReport
RQ1: source support (AlignScore and paired intervals)	—	—	—	—
RQ2: overlap and semantic quality (ROUGE and BERTScore)	—	—	—	—
RQ3: matched no-encoder, no-bridge, and no-grounded-copy ablations	—	—	—	—
RQ4: encoder-transfer matrix and quality/factuality diagnostics	—	—	—	—

direction-transformed, lower-is-better quantity for a hallucination-oriented view. This transform is not a calibrated probability. Claims about typed entity, number, or relation errors require a separately specified adjudicated audit.

Fill-in template after the locked runs: “Across PubMed, arXiv, BookSum, and GovReport, EviSeq obtained AlignScore consistency of *[TBD: dataset-specific values]*, compared with *[TBD: named matched decoder-only control(s)]*. The paired differences were *[TBD: differences]*, with 95% bootstrap intervals of *[TBD: intervals]*. The corresponding $H = 1 - C$ values were *[TBD: values]*. This pattern *[TBD: supports/does not support]* a change in the automatic source-support signal under this benchmark; it does not by itself establish a reduction in factual errors.”

RQ2: summary quality. RQ2 tests whether a change in source support is accompanied by acceptable summary coverage and semantic similarity. It compares EviSeq with the same decoder-only controls and T5Gemma2 on PubMed, arXiv, BookSum, and GovReport. The scorer-locked outcomes are ROUGE-1/2/L F1 and BERTScore-F1, read alongside length, truncation, empty-output, and repetition diagnostics.

Fill-in template after the locked runs: “Against each matched decoder-only control and T5Gemma2, EviSeq changed ROUGE-1, ROUGE-2, and ROUGE-L F1 on PubMed, arXiv, BookSum, and GovReport by *[TBD: dataset-by-baseline deltas]*. The paired 95% bootstrap intervals were *[TBD: intervals]*, and the corresponding BERTScore differences were *[TBD: differences]*. Output diagnostics showed *[TBD: length, truncation, empty-output, repetition, and copy findings]*. These results *[TBD: support/do not support]* a quality change under the registered protocol.” Here BERTScore measures semantic similarity; it is not a hallucination detector.

RQ3: component contribution. RQ3 compares the full EviSeq graph with separately trained no-encoder, no-AFMR-bridge, and no-grounded-copy controls. The decoder and training budget stay fixed across these runs on PubMed, arXiv, BookSum, and GovReport. Disabling a component in a finished checkpoint is only a diagnostic; the primary ablation comparison trains every row from the pretrained initialization. The interpretation remains local to the registered graph and protocol. Ablation differences can show which components contribute to the observed benchmark behavior, but they cannot establish universal causal necessity.

Fill-in template after the locked runs: “On PubMed, arXiv, BookSum, and GovReport, removing the encoder, AFMR bridge, and grounded copy route changed the pre-registered *[TBD: primary outcome(s)]* relative to full EviSeq by *[TBD: component-specific deltas]*. The paired intervals were *[TBD: intervals]*, and the associated ROUGE, AlignScore, and BERTScore diagnostics were *[TBD: values]*. The pattern *[TBD: does/does not]* indicate that *[TBD: component(s)]* contributes to the measured behavior under this registered setup.”

RQ4: encoder portability. RQ4 evaluates the same interface with PPLX-Embed v1 0.6B as the anchor and with Qwen3 causal encoder, Qwen3-Embedding-0.6B, and Nemotron-3-Embed-1B encoder variants on PubMed, arXiv, BookSum, and GovReport. For every variant, the transfer comparison will report encoder and decoder identifiers, hidden widths, trainable-parameter counts, and the same quality, factuality, and prediction diagnostics. Directionality and capacity belong to the encoder choice, so we will keep them explicit when interpreting transfer differences rather than attributing those differences to a single cause.

Fill-in template after the locked runs: “With the

decoder and protocol fixed, the PPLX-Embed v1 0.6B, Qwen3 causal encoder, Qwen3-Embedding-0.6B, and Nemotron-3-Embed-1B variants obtained [TBD: ROUGE-1/2/L values], [TBD: AlignScore values], and [TBD: BERTScore values] on PubMed, arXiv, BookSum, and GovReport. Their paired differences from the PPLX-Embed v1 0.6B anchor were [TBD: differences with intervals]. Latency, peak memory, length, repetition, and source-copy diagnostics were [TBD: diagnostics]. These results [TBD: support/do not support] portability across the tested encoder choices, conditional on the observed directionality and capacity differences.”

6 Discussion

The current result is an implementation boundary. The source tree provides an executable AFMR interface that retains a full retrieval tensor and an aligned final-state value tensor, separates refined retrieval keys from final-state values, and exposes a grounded copy route trained through the same gold-token cross-entropy objective as the language-model path. The tests support those execution properties, and the documented fixture smoke path provides a bounded way to exercise the complete runtime. These implementation checks do not answer whether the interface improves factuality, overlap quality, component-level behavior, or encoder portability. Answers to those questions depend on the held-out predictions and scorer-locked comparisons in Table 1.

An AlignScore increase, accompanied by a decrease in $H = 1 - C$ for $C = \text{AlignScore}_{\text{nli_sp}}$, would be consistent with greater source–summary consistency under the selected proxy. It would remain compatible with alternative explanations such as shorter outputs, stronger extractive behavior, truncation patterns, or dataset-specific reference properties. A manually audited factual-error subset, if available, would be needed to test whether the metric movement tracks unsupported entities, numbers, and relations. Copying a source token does not establish that the token has the correct role in a generated relation, so source-copy rate cannot replace that audit.

ROUGE and BERTScore answer a different question. A change in ROUGE-1/2/L F1 or BERTScore-F1 would describe reference overlap or semantic similarity under the locked preprocessing and scorer. It would not establish fac-

tual correctness. Length ratio, empty and truncated predictions, repetition, and copy diagnostics are needed to determine whether a metric difference reflects content selection, output constraints, or a change in source use. A factuality change with lower overlap would therefore be reported as a possible trade-off until the paired quality evidence is examined.

The RQ3 ablations provide component attribution only within the tested EviSeq graph. Removing a route changes the information path and may also change optimization dynamics or the number of trainable parameters. Separate train-from-pretrained runs, matched budgets, and validation-based development reduce these concerns but do not turn an ablation into a universal necessity claim. The narrow interpretation is whether each route contributes to the measured outcomes under the registered conditions.

The RQ4 matrix tests whether the interface remains usable when the encoder family, directionality, or capacity changes. A result that transfers to one variant would support portability to that tested choice. It would not isolate directionality from pretraining data, tokenizer behavior, hidden width, or parameter count. These factors should remain visible in the report so that a transfer effect is not attributed to the AFMR interface alone.

Several negative-space checks govern the eventual interpretation. A quality table without a corresponding prediction manifest would leave check-point and tokenizer provenance unresolved. An AlignScore value without its direction and paired uncertainty would not answer RQ1. A single aggregate across the four datasets would conceal a missing dataset-specific result. A reported ablation without a matched budget would not identify a component effect. A transfer claim without encoder identifiers and capacity diagnostics would not establish portability. The final paper should fill each of these evidence links before replacing the templates above with declarative results.

7 Limitations

First, the decisive quality evidence is absent at this stage. No four-dataset predictions or ROUGE-1.5.5, AlignScore, or BERTScore measurements are available for the proposed EviSeq configuration. The implementation tests cannot support a performance, factuality, or superiority claim.

Second, the present source tree does not in-

clude the final BookSum and GovReport preparation manifests or snapshots. Their source and target field choices, split identities, and preprocessing records must be supplied before the corresponding runs can be treated as reproducible evidence. Until then, Table 1 is a plan rather than a completed four-dataset evaluation.

Third, the outcome measures have limited scope. ROUGE measures reference overlap, and BERTScore measures semantic similarity; neither is a factuality test. AlignScore is a factuality proxy, and $1 - \text{AlignScore}$ is a direction transform rather than a calibrated hallucination probability. A manual factuality audit and multiple training seeds, if added, would address uncertainty that paired test-example intervals and a metric proxy leave open.

Fourth, source limits and conservative offset alignment can remove or reduce copy opportunities at truncation boundaries. Full-memory operation therefore means that all valid positions within the encoded prefix are retained; it does not guarantee that information outside the prefix is available or that the decoder uses every retained position correctly.

Fifth, encoder transfer is bounded by the selected families and by the confounding of directionality, pretraining, tokenization, width, and capacity. Latency and peak-memory costs also require measurement on the intended hardware. Tiny local fixtures establish runtime contracts, not deployment efficiency.

8 Conclusion

This paper defines EviSeq as a source-grounded full-memory interface for composing a pretrained encoder with a causal decoder. Its AFMR bridge keeps the encoded source available, adapts retrieval keys separately from final-state values, and supplies a shared source prior; the decoder adds a grounded copy route while training the complete graph with gold-token cross-entropy. These are testable design choices, not a measured quality claim.

The current evidence supports implementation consistency on controlled fixtures. It does not establish an AlignScore improvement, a ROUGE or BERTScore change, component necessity, or encoder portability. The next evidence required is the matched, four-dataset evaluation with scorer-locked predictions, paired uncertainty, ablations,

encoder-transfer runs, and the registered diagnostics. Until those runs are complete, EviSeq should be described as a reproducible interface and evaluation contract whose empirical contribution remains to be determined.

9 Reproducibility and Scope

The active implementation and proposed graph are the `src/eviseq_new` package with `architecture.name` set to `afmr_value_anchor`. The manuscript’s method name, model names, dataset names, and metric names are protected by the `\newcommand` definitions in the preamble so that later score insertion cannot silently change the comparison entities. The repository currently contains prepared pipelines for PubMed and arXiv; the BookSum and GovReport manifests, task recipes, and split-specific truncation statistics remain to be supplied before the four-dataset claims can be evaluated. The base YAML records the canonical AFMR defaults; any task-specific capacity or resource override must be retained in a resolved configuration and attached to its run. We will release the exact local model identifiers or immutable revisions, tokenizer identifiers, prompts, preprocessing reports, train and validation manifests, checkpoint names, generation settings, package versions, seeds, and failed-run records with the completed experiments.

All primary ROUGE values will use one Perl ROUGE 1.5.5 wrapper and one preprocessing path; the headline statistics are ROUGE-1/2/L F1 on the wrapper’s declared 0–100 scale. Python ROUGE 1.0.0 is retained only as a diagnostic and will not be mixed with the primary values. BERTScore-F1 is the headline semantic-similarity statistic, with precision and recall as secondary diagnostics. AlignScore is the automatic factual-consistency proxy: its native consistency is high-is-better, while the direction-transformed $H = 1 - C$ value, with $C = \text{AlignScore}_{\text{nli_sp}}$, is low-is-better and is not a calibrated hallucination probability. We will report paired bootstrap intervals over identical test IDs, dataset rows before a secondary macro statistic defined as the unweighted mean of the four dataset-level effects, and length, truncation, empty-output, repetition, copy, latency, and peak-memory diagnostics. Primary endpoints and exploratory contrasts will be labeled in the completed analysis.

10 Ethics and Data Use

The study uses publicly released summarization corpora and local pretrained checkpoints. Dataset licenses, access restrictions, and the provenance of each local checkpoint must be recorded in the experiment manifest before release. Summaries are model outputs and may contain unsupported biomedical, narrative, or policy statements; automatic factuality scores are screening signals rather than medical, legal, or policy advice. We do not claim that the interface guarantees factual correctness.

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Component	Locked default
Encoder	PPLX-Embed v1 0.6B
Decoder	Qwen3-0.6B
AFMR controller	256 dimensions
Depth readout	Four final taps, rank 128 residual
Feature readout	Rank 256 residual, feature-wise gate
Source prior	Windows 32/128/512, 50% overlap
Cross-attention	Copied block in every decoder layer
Copy route	Offset-aligned source-token mixture
Objective	Gold-token mixture cross-entropy only
Precision	FP32 state with CUDA BF16 autocast
Decoding	One beam, greedy, sampling disabled

Table 2: The EviSeq configuration authority is the `src/eviseq_new afmr_value_anchor` graph. The table lists the base-profile defaults; each task recipe’s resolved architecture, model paths, data paths, lengths, and resource settings will be archived with its run. Score fields are intentionally absent until the registered runs finish.

Yanzhao Zhang, Mingxin Li, Dingkun Long, Xin Zhang, Huan Lin, Baosong Yang, Pengjun Xie, An Yang, Dayiheng Liu, Junyang Lin, Fei Huang, and Jingren Zhou. 2025b. Qwen3 Embedding: Advancing text embedding and reranking through foundation models. *arXiv preprint arXiv:2506.05176*.

A Locked Configuration Summary

B Score-Filling Checklist

Before replacing any *[TBD:]* cell, verify that the prediction manifest records the same dataset split, example IDs, tokenizer, checkpoint rule (`last.pt`), preprocessing, generation parameters, and scorer version for every row in that table. A completed cell must be accompanied by a prediction file, metric JSON, and paired uncertainty estimate. If a run fails or uses a different protocol, retain *[TBD:]* and report the omission rather than substituting a convenient score.